Lecture 12

Bell's theorem: no local hidden variables

Objectives. *The electron and the proton of hydrogen's ground state ψ^- give opposite readings along every common axis. Could each reading have been fixed in advance, by some property of the atom that quantum mechanics does not describe? This lecture states what such a* local hidden-variable model *is, and proves the CHSH inequality. In every such model, a certain combination S of four correlations satisfies $|S| \le 2$. The singlet gives $|S| = 2\sqrt{2}$. No local model reproduces the predictions of quantum mechanics, and experiments agree with quantum mechanics. Three spins give a sharper conflict, with certainties in place of averages.*

12.1 Perfect anticorrelation

Let hydrogen's two spins be in the singlet $\psi^- = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ of Lecture 10. Read the electron's spin along a unit vector $\mathbf{a}$ and the proton's along a unit vector $\mathbf{b}$. The two observables $\mathbf{a}\cdot\boldsymbol{\sigma}\otimes\mathbb{1}$ and $\mathbb{1}\otimes\mathbf{b}\cdot\boldsymbol{\sigma}$ commute, so by Theorem 5.2 the two readings can be made at once, and each reading is $+1$ or -1. The mean of the product of the two readings is their *correlation*,

$$E(\mathbf{a},\mathbf{b}) = \langle \psi^-, (\mathbf{a}\cdot\boldsymbol{\sigma}\otimes\mathbf{b}\cdot\boldsymbol{\sigma})\,\psi^-\rangle\,,$$

since $\mathbf{a}\cdot\boldsymbol{\sigma}\otimes\mathbf{b}\cdot\boldsymbol{\sigma} = \sum_{\alpha,\beta}\alpha\beta\,E_\alpha\otimes G_\beta$, where E_α and G_β are the projections of the readings $\alpha,\beta = \pm 1$. The correlation is a mean, not a projection. We keep the letter E for it, as is usual.

Proposition 12.1 (The singlet's correlations). *For all unit vectors* $\mathbf{a}$ *and* $\mathbf{b}$*,*

$$E(\mathbf{a},\mathbf{b}) = -\mathbf{a}\cdot\mathbf{b}.$$

Proof. First, for each $c = x, y, z$,

$$(\sigma_c\otimes\mathbb{1} + \mathbb{1}\otimes\sigma_c)\,\psi^- = 0. \tag{12.1}$$

For $c = z$: $(\sigma_z\otimes\mathbb{1})\psi^- = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ and $(\mathbb{1}\otimes\sigma_z)\psi^- = \frac{1}{\sqrt{2}}(-|01\rangle - |10\rangle)$. For $c = x$: $(\sigma_x\otimes\mathbb{1})\psi^- = \frac{1}{\sqrt{2}}(|11\rangle - |00\rangle)$ and $(\mathbb{1}\otimes\sigma_x)\psi^- = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$. For $c = y$ the two terms are $\frac{i}{\sqrt{2}}(|11\rangle + |00\rangle)$ and $-\frac{i}{\sqrt{2}}(|00\rangle + |11\rangle)$. Next, by (12.1) for the index d,

$$\langle\psi^-, (\sigma_c\otimes\sigma_d)\psi^-\rangle = \langle\psi^-, (\sigma_c\otimes\mathbb{1})(\mathbb{1}\otimes\sigma_d)\psi^-\rangle = -\langle\psi^-, (\sigma_c\sigma_d\otimes\mathbb{1})\psi^-\rangle = -\operatorname{Tr}(\rho_1\sigma_c\sigma_d),$$

where $\rho_1 = \frac{1}{2}\mathbb{1}$ is the electron's reduced density matrix (§ 11.2). By the Pauli rule, $\frac{1}{2}\operatorname{Tr}(\sigma_c\sigma_d)$ is 1 for $c = d$ and 0 otherwise. So $\langle\psi^-, (\sigma_c\otimes\sigma_d)\psi^-\rangle = -\delta_{cd}$, and expanding $\mathbf{a}\cdot\boldsymbol{\sigma} = \sum_c a_c\sigma_c$ and $\mathbf{b}\cdot\boldsymbol{\sigma}$ gives $E(\mathbf{a},\mathbf{b}) = -\sum_c a_c b_c$. ∎

The two readings are equal with probability $\frac{1}{2}(1+E) = \frac{1}{2}(1 - \mathbf{a}\cdot\mathbf{b})$. Along a common axis, $\mathbf{b} = \mathbf{a}$, they are always opposite, since $E(\mathbf{a},\mathbf{a}) = -1$. Lecture 10 found this along z and x. It holds along every axis.

Einstein, Podolsky and Rosen drew a conclusion from such perfect correlations in 1935, in a different example. Bohm put it in terms of two spins. A reading of the proton along $\mathbf{a}$ predicts with certainty what a reading of the electron along $\mathbf{a}$ will give. They added a premise: a reading made on the proton, far away, does not disturb the electron's physical situation. From the two they concluded that the electron's reading along $\mathbf{a}$ must have been fixed before it was made, for every axis $\mathbf{a}$, by something the state ψ^- does not record. Their premise is stronger than Proposition 11.8, which says only that the electron's pooled statistics do not change. Bell asked in 1964 whether such a description can reproduce the predictions of quantum mechanics. This lecture shows that it cannot.

12.2 Local models

Two observers, one at each spin, each choose one of two settings: an axis $\mathbf{a}_x$, $x = 1, 2$, for the electron, and an axis $\mathbf{b}_y$, $y = 1, 2$, for the proton. Each records a reading, $\alpha = \pm 1$ and $\beta = \pm 1$. A model of the experiment that fixes the readings in advance, or at least their probabilities, has the following form.

Definition 12.2 (Local model)**.** A *local model* consists of

1. a finite set Λ of *hidden states* λ, with probabilities $p(\lambda) \geq 0$, $\sum_\lambda p(\lambda) = 1$, that do not depend on the settings x and y;
2. for each λ and each setting x, probabilities $P_1(\alpha \mid x, \lambda)$ of the electron's readings $\alpha = \pm 1$, and for each λ and each setting y, probabilities $P_2(\beta \mid y, \lambda)$ of the proton's readings $\beta = \pm 1$;

such that the readings α and β occur together, at the settings x and y, with probability

$$\mathrm{Prob}(\alpha, \beta \mid x, y) = \sum_{\lambda \in \Lambda} p(\lambda)\, P_1(\alpha \mid x, \lambda)\, P_2(\beta \mid y, \lambda). \tag{12.2}$$

Each hypothesis has a meaning. The hidden state λ is prepared with the pair, before the settings are chosen. That its probabilities do not depend on the settings is an assumption. Given λ, the electron's reading depends on the electron's setting only, and not on the proton's setting or reading. This is *locality*. The two readings can be made far apart, at the same time. The model may still correlate the readings, through λ.

In a *deterministic* local model every $P_1(\alpha \mid x, \lambda)$ and $P_2(\beta \mid y, \lambda)$ is 0 or 1. Each reading is then fixed by λ and the setting. This is the case "the readings were there all along". The definition allows more, and the theorem below covers all of it.

For each λ put the *local means*

$$\bar{A}_x(\lambda) = P_1(+1 \mid x, \lambda) - P_1(-1 \mid x, \lambda), \qquad \bar{B}_y(\lambda) = P_2(+1 \mid y, \lambda) - P_2(-1 \mid y, \lambda),$$

numbers between -1 and 1. By (12.2), the correlation at the settings x, y is

$$\begin{aligned} E_{xy} &= \sum_{\alpha,\beta} \alpha\beta\ \mathrm{Prob}(\alpha, \beta \mid x, y) = \sum_\lambda p(\lambda) \Big(\sum_\alpha \alpha P_1(\alpha \mid x, \lambda) \Big) \Big(\sum_\beta \beta P_2(\beta \mid y, \lambda) \Big) \\ &= \sum_\lambda p(\lambda)\, \bar{A}_x(\lambda) \bar{B}_y(\lambda). \end{aligned} \tag{12.3}$$

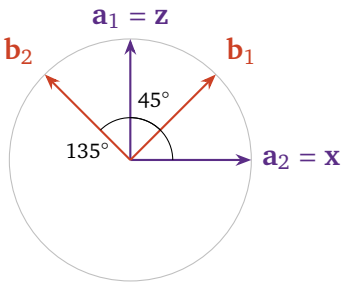


Figure 12.1: The four axes of Example 12.5, in the xz-plane of the Bloch sphere: $\mathbf{b}_1$ and $\mathbf{b}_2$ bisect the angles between $\mathbf{z}$ and $\pm\mathbf{x}$. Three of the pairs in S are 45° apart, and the pair $\mathbf{a}_2$, $\mathbf{b}_2$ is 135° apart.

Remark 12.3 (More hidden states)**.** A hidden state could range over a continuum, with a probability density in place of $p(\lambda)$. The sums over λ then become integrals, and the proof below is unchanged. We keep Λ finite, so that no integration theory is needed.

12.3 The CHSH inequality

Clauser, Horne, Shimony and Holt (CHSH) combined the four correlations into

$$S = E_{11} + E_{12} + E_{21} - E_{22}. \tag{12.4}$$

Theorem 12.4 (CHSH)**.** *In every local model,* $|S| \le 2$.

Proof. First, for real numbers u and v,

$$|u+v| + |u-v| = 2\max(|u|, |v|).$$

Indeed, if $|u| \ge |v|$, the numbers $u+v$ and $u-v$ each have the sign of u or vanish, so $|u+v| + |u-v| = |(u+v)+(u-v)| = 2|u|$. The case $|v| \ge |u|$ is the same with u and v exchanged. Next, fix λ, and put $u = \bar{B}_1(\lambda)$, $v = \bar{B}_2(\lambda)$. Since $\left|\bar{A}_x(\lambda)\right| \le 1$,

$$\left|\bar{A}_1(u+v) + \bar{A}_2(u-v)\right| \le |u+v| + |u-v| = 2\max(|u|, |v|) \le 2.$$

Finally, by (12.3),

$$S = \sum_{\lambda} p(\lambda)\Big(\bar{A}_1(\lambda)\big(\bar{B}_1(\lambda) + \bar{B}_2(\lambda)\big) + \bar{A}_2(\lambda)\big(\bar{B}_1(\lambda) - \bar{B}_2(\lambda)\big)\Big),$$

an average of numbers between -2 and 2, so $|S| \le 2$. ∎

The bound is reached. In the deterministic model with one hidden state and $\bar{A}_1 = \bar{A}_2 = \bar{B}_1 = \bar{B}_2 = 1$, every correlation is 1, so $S = 1 + 1 + 1 - 1 = 2$.

12.4 The singlet violates it

Example 12.5 (Four settings for the singlet)**.** Figure 12.1 shows the settings

$$\mathbf{a}_1 = \mathbf{z}, \qquad \mathbf{a}_2 = \mathbf{x}, \qquad \mathbf{b}_1 = \tfrac{1}{\sqrt{2}}(\mathbf{z}+\mathbf{x}), \qquad \mathbf{b}_2 = \tfrac{1}{\sqrt{2}}(\mathbf{z}-\mathbf{x}).$$

1. *The correlations.* By Proposition 12.1, $E_{xy} = -\mathbf{a}_x \cdot \mathbf{b}_y$. The three products $\mathbf{a}_1 \cdot \mathbf{b}_1$, $\mathbf{a}_1 \cdot \mathbf{b}_2$, $\mathbf{a}_2 \cdot \mathbf{b}_1$ are $\frac{1}{\sqrt{2}}$, and $\mathbf{a}_2 \cdot \mathbf{b}_2 = -\frac{1}{\sqrt{2}}$. So $E_{11} = E_{12} = E_{21} = -\frac{1}{\sqrt{2}}$ and $E_{22} = \frac{1}{\sqrt{2}}$.
2. *The sum.* $S = -\frac{3}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -2\sqrt{2}$, so $|S| = 2\sqrt{2} \approx 2.83 > 2$.

By Theorem 12.4, no local model gives these four correlations. This is *Bell's theorem*, in the form of Clauser, Horne, Shimony and Holt: no local model reproduces the predictions of quantum mechanics for the singlet. The perfect anticorrelation of § 12.1 alone does not decide the question. A local model can give $E(\mathbf{a}, \mathbf{a}) = -1$ for every axis, as Exercise Sheet 12 shows. It is the correlations at the angles in between that no local model can match.

Entanglement is needed for this.

Proposition 12.6 (Products have local models)**.** *Let a pair of qubits be an ensemble $\{(p_k, u_k \otimes w_k)\}$ of product states, with unit vectors u_k and w_k, and let any two observables with readings ± 1 be read on each side. Then $|S| \leq 2$.*

Proof. Let $E_\alpha^{(x)}$ and $G_\beta^{(y)}$ be the projections of the readings at the settings x and y. By Theorem 3.4, the readings α and β occur with probability

$$\sum_k p_k \left\langle u_k \otimes w_k, (E_\alpha^{(x)} \otimes G_\beta^{(y)})(u_k \otimes w_k) \right\rangle = \sum_k p_k \left\langle u_k, E_\alpha^{(x)} u_k \right\rangle \left\langle w_k, G_\beta^{(y)} w_k \right\rangle .$$

This is (12.2) with the hidden states k, the probabilities p_k and the response probabilities $\left\langle u_k, E_\alpha^{(x)} u_k \right\rangle$ and $\left\langle w_k, G_\beta^{(y)} w_k \right\rangle$. By Theorem 12.4, $|S| \leq 2$. ∎

So a violation of the CHSH inequality rules out every ensemble of product states. The correlated mixture of § 10.4 obeys the bound, and ψ^- does not.

12.5 Three spins: certainty instead of statistics

The CHSH inequality compares averages. With three spins the conflict is sharper. It concerns predictions that quantum mechanics makes with certainty. By Postulate 10.1, applied twice, three spins have the state space $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$, with the basis $|jkl\rangle = |j\rangle \otimes |k\rangle \otimes |l\rangle$, and a reading of the second spin alone is a reading of $\mathbb{1} \otimes A \otimes \mathbb{1}$. Take the state

$$\psi_{\mathrm{GHZ}} = \tfrac{1}{\sqrt{2}}\big(|000\rangle + |111\rangle\big),$$

named after Greenberger, Horne and Zeilinger. The argument below is in the form Mermin gave it. Each spin is read along x or along y.

Proposition 12.7 (Four certain products)**.** *ψ_{GHZ} is an eigenvector of $\sigma_x \otimes \sigma_x \otimes \sigma_x$ with the eigenvalue $+1$, and of $\sigma_x \otimes \sigma_y \otimes \sigma_y$, $\sigma_y \otimes \sigma_x \otimes \sigma_y$ and $\sigma_y \otimes \sigma_y \otimes \sigma_x$ with the eigenvalue -1.*

Proof. Since $\sigma_x |0\rangle = |1\rangle$, $\sigma_y |0\rangle = \mathrm{i}\,|1\rangle$ and $\sigma_y |1\rangle = -\mathrm{i}\,|0\rangle$, each of the four matrices takes $|000\rangle$ to $|111\rangle$ and $|111\rangle$ to $|000\rangle$, times a factor: 1 for the first, and $\mathrm{i} \cdot \mathrm{i} = (-\mathrm{i})(-\mathrm{i}) = -1$ for the other three. ∎

In each of the four settings the three readings are of observables acting on different factors, so they commute and are read at once (Lecture 5). The product of the three readings is then certain: $+1$ when all three spins are read along x, and -1 when one is read along x and two along y.

A local model of three spins is Definition 12.2 with a third side: hidden states λ with probabilities $p(\lambda)$ that do not depend on the settings, and response probabilities $P_k(\alpha_k \mid s_k, \lambda)$ of the reading $\alpha_k = \pm 1$ of spin k at its setting s_k (here σ_x or σ_y), such that

$$\text{Prob}(\alpha_1, \alpha_2, \alpha_3 \mid s_1, s_2, s_3) = \sum_\lambda p(\lambda)\, P_1(\alpha_1 \mid s_1, \lambda)\, P_2(\alpha_2 \mid s_2, \lambda)\, P_3(\alpha_3 \mid s_3, \lambda). \tag{12.5}$$

As in (12.3), the mean of the product of the three readings is

$$\sum_\lambda p(\lambda)\, \bar{A}_1(s_1, \lambda) \bar{A}_2(s_2, \lambda) \bar{A}_3(s_3, \lambda),$$

with the local means $\bar{A}_k(s_k, \lambda)$ between -1 and 1.

Lemma 12.8 (Certainty forces values). *If in a local model of three spins the product of the three readings at some settings has the mean ± 1, then for every λ with $p(\lambda) > 0$ the three local means at these settings are ± 1.*

Proof. A mean of numbers between -1 and 1, with weights $p(\lambda)$, is ± 1 only if every number with positive weight is ± 1. A product of three numbers between -1 and 1 has modulus 1 only if each of them has. ∎

Perfect correlations force values. For each such λ the readings at these settings are fixed, as Einstein, Podolsky and Rosen argued.

Theorem 12.9 (No local model of the GHZ state). *No local model of three spins gives the four certain products of Proposition 12.7.*

Proof. Suppose one does. Each spin's setting σ_x occurs in a certain product, and so does its setting σ_y. By Lemma 12.8, every λ with $p(\lambda) > 0$ fixes the values $x_k(\lambda) = \pm 1$ and $y_k(\lambda) = \pm 1$ of the readings of spin k along x and along y (values, not settings). A local mean depends on the spin's own setting only, so each value is the same in every product in which it occurs, and

$$x_1 x_2 x_3 = 1, \qquad x_1 y_2 y_3 = y_1 x_2 y_3 = y_1 y_2 x_3 = -1.$$

The product of the last three is $x_1 x_2 x_3\, (y_1 y_2 y_3)^2 = x_1 x_2 x_3 = 1$, not $(-1)^3 = -1$. ∎

No inequality is needed. The contradiction is with four predictions of probability one. Real correlations are never perfect, so experiments test an inequality instead (Mermin's, Exercise Sheet 12). The contradiction needs all three spins. Each pair has the reduced state $\frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$ (Lecture 11), the state of an ensemble of products, so no pair alone violates the CHSH inequality (Proposition 12.6). Moreover, the two readings σ_x and σ_y of one spin do not commute, so no run reads both.

12.6 What the violation does and does not say

The violation allows no signalling. Whatever axis is chosen for the electron, and as long as nobody conditions on its outcome, the proton reads +1 along every axis with probability $\text{Tr}(\rho_2 G_+) = \frac{1}{2}$, by Proposition 11.8 and $\rho_2 = \frac{1}{2}\mathbb{1}$. The choice of setting on one side cannot be read off the other side's readings alone. The correlations show only when the two records are brought together. "Nonlocal" in this sense means that no local model exists. It does not mean an effect that travels faster than light.

The theorem has three hypotheses, and an experiment must justify each. First, the factorised law of Definition 12.2: given λ, the reading on one side depends on neither the setting nor the reading on the other side. Experiments choose each setting at a time and place from which no signal at the speed of light could reach the other side's reading in time. This rules out such signals. It does not by itself prove the factorised law. Second, every run must give a reading ± 1 and be counted. If many runs are lost, the counted ones need not obey the inequality. Third, the settings must be independent of the hidden state, even when the hidden state is prepared earlier. No experiment can prove this, and it remains an assumption.

Quantum mechanics does not give every value of S either. For every state of two qubits and every four settings, $|S| \leq 2\sqrt{2}$ (Tsirelson's bound, Exercise Sheet 12). The singlet with the settings of Example 12.5 reaches it.

The physics

The first tests were made with pairs of photons, whose polarisation is a qubit. Freedman and Clauser (1972) measured the polarisation correlations of photons emitted in pairs by calcium atoms and found them in agreement with quantum mechanics and in violation of a Bell inequality, under auxiliary assumptions. Aspect, Dalibard and Roger (1982) switched the analysers' settings while the photons were in flight. A polariser at the angle ϑ reads the polarisation along a Bloch axis at the angle 2ϑ, so the Bloch angles of Example 12.5, 45° apart, are polariser angles 22.5° apart (Lecture 5). In 2015 three groups closed the loopholes of locality and of lost runs in one experiment each: Hensen and co-workers with electron spins in diamond, using the CHSH form, and Giustina and co-workers and Shalm and co-workers with photons, using a related inequality of Clauser and Horne. Aspect, Clauser and Zeilinger received the Nobel Prize in Physics in 2022 for such experiments. Pan, Bouwmeester, Daniell, Weinfurter and Zeilinger (2000) prepared three photons in the GHZ state and found the correlations of § 12.5, imperfect as every count is, in agreement with quantum mechanics.

Bell's theorem

The singlet: $E(\mathbf{a}, \mathbf{b}) = -\mathbf{a} \cdot \mathbf{b}$ (Proposition 12.1). *Local models:* a hidden state with setting-independent probabilities, and local response probabilities; then $E_{xy} = \sum_\lambda p(\lambda) \bar{A}_x(\lambda) \bar{B}_y(\lambda)$ (Definition 12.2). *CHSH:* $S = E_{11} + E_{12} + E_{21} - E_{22}$ has $|S| \leq 2$ in every local model (Theorem 12.4); the singlet gives $S = -2\sqrt{2}$ (Example 12.5). *GHZ:* $\psi_{\mathrm{GHZ}} = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$; products $xxx = +1$, $xyy = yxy = yyx = -1$, certain (Theorem 12.9).

Summary.

- *The singlet's correlations*: $E(\mathbf{a}, \mathbf{b}) = -\mathbf{a} \cdot \mathbf{b}$, perfectly opposite along every common axis (Proposition 12.1).
- *CHSH*: every local model has $|S| \leq 2$ (Theorem 12.4).
- *Bell's theorem*: the singlet gives $|S| = 2\sqrt{2}$, so no local model reproduces quantum mechanics (Example 12.5).
- *Three spins*: no local model gives the four certain products of ψ_{GHZ} (Theorem 12.9).

Lecture 13 returns to the two spins of hydrogen and asks how they rotate together: which total spins two spins can make.

References

Nielsen and Chuang, *Quantum Computation and Quantum Information*, §2.6, pp. 111–117. Sakurai and Napolitano, *Modern Quantum Mechanics*, §3.10, pp. 224–231. Bell, *Physics* 1, 195 (1964). Clauser, Horne, Shimony and Holt, *Physical Review Letters* 23, 880 (1969). Aspect, Dalibard and Roger, *Physical Review Letters* 49, 1804 (1982). Hensen et al., *Nature* 526, 682 (2015). Giustina et al. and Shalm et al., *Physical Review Letters* 115, 250401 and 250402 (2015). Greenberger, Horne and Zeilinger, in *Bell's Theorem, Quantum Theory and Conceptions of the Universe* (Kluwer, 1989), pp. 69–72. Mermin, *American Journal of Physics* 58, 731 (1990). Pan et al., *Nature* 403, 515 (2000).